

SECTION A

ITEM 1

- (a) You are an electrical engineering technician working on a smart city initiative. Your team is installing automatic street lights in a new residential area to save energy. The system relies on a photocell that automatically turns the street lights on when the sunsets and off when it rises.

A local resident, however complains that the lights flicker during cloudy days and wonders how a simple light sensor can make such quick decisions. Your manager tasks you with explaining the principle of the device to the public and providing a brief on how this photoelectric effect is used in physics laboratory to determine a fundamental constant of nature.

Task.

As a physics consultant;

- (i) Help to explain briefly how the photocell in the street light uses the photoelectric effect to detect darkness.
 - (ii) Identify two other daily life applications of the photoelectric effect and explain how one of them operates
 - (iii) Sketch a labelled diagram of the apparatus required to determine Planck's constant.
 - (iv) List the measurements taken during the experiment. State the equation connecting stopping potential (V_s), frequency, f , and Planck's constant, h . Explain how a graph of these measurements is used to determine the value of h .
- (b) You are working as a Junior researcher in an environmental physics laboratory studying the charge distribution of aerosolized microscopic oil droplets used in an industrial air cleaner. The efficiency of the filter depends on knowing how many electrons are on each tiny oil droplet but the droplets acquire different charges through friction during spraying.

Task.

As a physics student, you have successfully used an atomizer to spray oil, ionized the air with x-rays and used a microscope to observe different droplets.

- (i) Use the modern Millikan's oil drop apparatus to show how the electronic charge of an oil drop can be determined.
- (ii) In an experiment used in (i) above, a charged oil drop of radius $9.2 \times 10^{-7} \text{ m}$ and density 800 kg m^{-3} is held stationary in an electric field of intensity $4.0 \times 10^4 \text{ V m}^{-1}$. How many charges are on the drop and what is the electric field intensity then be applied vertically to move oil drop with a velocity of 0.005 ms^{-1} upwards.

Hint:

Density of air = 1.29 kg m^{-3}

Co-efficient of viscosity of air = $1.8 \times 10^{-5} \text{ N s m}^{-1}$

SECTION B

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ITEM 2

- ✓ (a) During a physics field trip to a local amusement park, students are analyzing the swing Ride (chair- o -plane). The ride consists of a central rotating tower with seats attached to chains. When the ride operates at a constant high speed, the seats rise and move in a horizontal circle forming a conical pendulum. A student calculates the total mass of seat and passenger 80kg, length of the chain 10m and angle of the chain with the vertical 30° . Assuming that the chain is light, inextensible and air resistance is negligible.

Task.

Using the information above;

- (i) Draw a free body diagram showing the forces acting on the seat (passenger) system.
- (ii) Calculate the tension in the chain and the constant linear speed of the seat.

- ✓ (b) A security system in a high technical ware house involves a 0.5kg surveillance camera mounted at the edge of a loading dock 2.0meters above the concrete floor. A courier, running to deliver a package, accidentally throws a 2.0kg metal briefcase horizontally at 10ms^{-1} . The briefcase strikes the camera and the two objects stick together forming a combined mass. The collision causes the combined object to fall off the loading dock.

Hint: Assuming friction is negligible

Task.

As a physics student, how would you help the security personnel to;

- (i) Calculate the horizontal velocity of the combined camera briefcase system immediately after collision.
- (ii) Determine the total horizontal distance the combined object travels from the edge of the dock before hitting the floor.

- ✓ (c) A worker is trying to clean a window 4metres above the ground. They have a 5m long uniform ladder weighing 200N.

The ground is rough (coefficient of friction $\mu = 0.4$) but the wall is smooth. The worker weighing 800N wants to know if they can safely reach the window without the ladder slipping.

Task.

- (i) ✓ The ladder is placed so that its base is 3m from the wall. Calculate the angle the ladder makes with the ground. At what maximum distance up the ladder can the 800N worker stand before the ladder is on the verge of slipping.
- (ii) To make the ladder safer, worker asks a friend to stand at the base of the ladder, pushing it against the wall with a constant horizontally force of 100N while the worker is at the top. If the worker is now 4m up, calculate the new total friction needed at the base of the ladder to prevent slipping.

SECTION C

ITEM 3

- (a) Sarah an s.6 student visiting her aunt who is a doctor observes a high tech. medical procedure. The doctor uses an endoscope to examine a patient's stomach. Sarah notices the endoscope is a flexible tube that lights up the inside of the body and sends a clear image back to screen. During the exam, a tiny camera at the trip of the endoscope captures images which are transferred to a monitor. Sarah remembers studying that these cameras use convex lenses to focus light.

The patient's data is sent to a specialist in another city using fiber optic internet.

The medical instruments are sterilized using a UV light box.

Task.

- (i) Explain how light travels through the fiber optic cables in the endoscope to illuminate the patient's stomach and return an image, mentioning the critical angle.
 - (ii) The endoscope camera uses a convex lens to capture images. If the camera lens has a focal length of 5mm and is used to take a picture if a tumor 15mm away, calculate the image distance from the lens and state the nature of the image.
 - (iii) Identify the wave phenomenon that allows the UV light box to kill bacteria and explain why the UV light is more effective than a visible light in this context.
 - (iv) If the doctor is shortsighted and wears glasses to view the monitor, what type of lens is in their spectacles and how does it correct their vision?
- (b) It was a bright Saturday morning when Annah and Moses prepared for their long-awaited wedding. The ceremony was set to take place in White Horse In. The decorators were busy setting strings of decorative lights powered by electricity being produced from the generator. As the band began playing some soft music, people at the main market couldn't get the sound clearly.

Task.

- (i) Explain briefly how the sound produced from the band group were able to reach people at different locations.
- (ii) Why most of the musical instruments being blown by the band group were open at both ends.
- (iii) Explain briefly why the sound detected at the main market wasn't clear in the morning and generating during the day time than at night
- (iv) State the wave form when the guitar was being played and derive its equation at the wedding Venue, the loudspeaker produces a musical note with frequency of 440Hz. If the speed of the sound in air is 340ms^{-1} .
- (v) Determine the frequency of the sound detected by a boda- boda man riding at 60kmhr^{-1} and he is moving towards the venue.

ITEM 4

You are helping organize the school sports day in a certain school. To serve hot tea to students 2.5kg of water is boiled and poured into a portable steel dispenser of thickness 2.5mm. The sides and base of the dispenser are lined with an 8mm thick layer of polystyrene foam to reduce heat loss to the surrounding air. The total surface area through which heat can escape by conduction is 0.8m^2 and the tea inside stays at approximately 80°C .

Hint.

- ❖ Thermo-conductivity of steel = $50\text{Wm}^{-1}\text{K}^{-1}$
- ❖ Thermo-conductivity of polystyrene foam = $0.035\text{Wm}^{-1}\text{K}^{-1}$
- ❖ Temperature of the surrounding air = 25°C

Task

- (i) Briefly explain why polystyrene foam is chosen for this purpose compared to a good conductor like steel.
- (ii) Describe an experiment to determine the thermal conductivity of a sample of the same polystyrene foam, stating all the necessary precautions.
- (iii) Calculate the rate of loss of heat through the polystyrene foam insulation.
- (iv) If the dispenser was made with no insulation layer (just bare steel), explain what would happen to the temperature of the tea overtime compared to the insulated version.

SECTION D

ITEM 5

An Engineering trainee working on a communication tower with small charged components. He observed a faint bluish glow and heard a crackling sound around conductors carrying 220kV especially during humid nights. The trainee did not know what was happening to the wires and thought that they have decayed. He later observed that the glow was stronger near the sharp transmission line fittings.

During a laboratory experiment, small charged spheres are placed at the corners of a square with sides of 20cm. The charges are;

- Charge at corners A = $+3\mu\text{C}$
- Charge at corner B = $-2\mu\text{C}$
- Charge at corner C = $+4\mu\text{C}$
- Charge at corner D = $+7\mu\text{C}$

The trainee would wish to know the electrostatic force and potential at corner D due to the charges at other corners such that he can make informed decisions. He did not know how to calculate for that force and potential.

Task.

As a physics student, help the trainee to;

- (i) Explain observations around conductors carrying 220kV
- (ii) Explain the disadvantages of what was observed on conductors and how he can minimize them.
- (iii) Calculate the force and potential at corner D.

ITEM 6

(a) You are a Junior engineer at a material testing laboratory. Your supervisor has provided a sample of a newly developed alloy wire and requested you to determine its electrical resistivity ρ , to assess its suitability for high efficiency heating elements.

You have access to a D.C power supply, an ammeter, a voltmeter, a micrometer screw gauge, a meter rule and connecting wires.

Task.

- (i) Sketch a labelled circuit diagram showing how you would connect the wire in series with an ammeter, power supply and a voltmeter in parallel across a segment of the wire.
- (ii) Describe the step-by-step procedures to measure the diameter of the wire, length of the wire segment and the corresponding voltage and current.
- (iii) Describe how to plot a graph of resistance R against length L and use it to determine the resistivity of alloy wire.

(b) An audio Engineer is designing a simple R-C high pass filter for a speaker system.

They are using a $1.0\text{k}\Omega$ resistor in series with a 100nF capacitor connected to an A.C signal generator. To ensure the circuit behaves correctly, the engineer must determine impedance of the circuit at a signal frequency of 1kHz and maximum Energy stored in the capacitor if the peak voltage across it is measured at 5V

Task.

- (i) As a student, help the audio engineer to determine the impedance of the series circuit and Energy stored in the capacitor
- (ii) Describe with the aid of a labeled diagram, structure and action of a hot wire ammeter as a device that is used to measure alternating current.

END